Homework 1

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Math 23
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1 1.2

Question 10. A radioactive material, such as the isotope thorium-234, disintegrates at a rate proportional to the amount currently present. If $Q(t)$ is the amount present at time t, then $dQ/dt = -rQ$, where $r > 0$ is the decay rate.

(a) If 100 mg of thorium-234 decays to 82.04 mg in 1 week, determine the decay rate r.

(b) Find an expression for the amount of thorium-234 present at any time t.

(c) Find the time required for the thorium-234 to decay to one-half its original amount.

Solution. Given the first-order differential equation, we use separation of variables as such:

$$\frac{dQ}{Q} = -rdt \implies \int_{100\text{ mg}}^{82.04\text{ mg}} \frac{dQ}{Q} = -\int_0^{1\text{ week}} rdt$$

Evaluating both definite integrals gives

$$\ln|82.04\text{ mg}| - \ln|100\text{ mg}| = -r(1\text{ week}).$$

Using logarithm rules and algebraic rules, we get

$$\boxed{r = \frac{\ln(100/82.04)}{1\text{ week}} \approx 0.198\text{ week}^{-1}.}$$

For part (b) to solve the differential equation as such:

$$\begin{aligned}
\frac{dQ}{Q} &= -rdt \\
\int_{Q_0}^{Q(t)} \frac{dQ'}{Q'} &= -r\int_0^t dt' \\
\left[\ln|Q'|\right]_{Q_0}^{Q(t)} &= -r\left[t'\right]_0^t \\
\ln|Q(t)| - \ln|Q_0| = \ln\left|\frac{Q(t)}{Q_0}\right| &= -rt \\
\frac{Q(t)}{Q_0} &= e^{-rt} \implies \boxed{Q(t) = Q_0 e^{-\left(0.198\text{ week}^{-1}\right)t}}
\end{aligned}$$

For part (c), when a radioactive material reaches its half-life, it is true that $Q(t)/Q_0 = 1/2$. Therefore, using one of our previous equations,

$$\ln\left|\frac{Q(t)}{Q_0}\right| = \ln\left|\frac{1}{2}\right| = -\left(0.198\ \text{week}^{-1}\right)t$$

$$\boxed{t = \frac{\ln|2|}{0.198\ \text{week}^{-1}} \approx 3.5\ \text{weeks}}$$

$\square$

2 1.3

Question 8. Verify that each given function is a solution of the differential equation

$$y'''' + 4y''' + 3y = t.$$

- $y_1(t) = t/3$
- $y_2(t) = e^{-t} + t/3$

Solution. For $y_1(t)$, it is true that $y_1' = 1/3$ and the subsequent derivatives are zero. Therefore,

$$\boxed{0 + 4(0) + 3\left(\frac{t}{3}\right) = t.}$$

Similarly, for $y_2(t)$, is is true that $y_2''' = -e^{-t}$ and $y_2'''' = e^{-t}$. Therefore,

$$\boxed{e^{-t} + 4\left(-e^{-t}\right) + 3\left(e^{-t} + t/3\right) = \cancel{-4e^{-t} + 4e^{-t}} + t = t.}$$

$\square$

Question 10. Verify that the given function is a solution of the differential equation

$$y' - 2ty = 1.$$

- $y = e^{t^2} \int_0^t e^{-s^2}\, ds + e^{t^2}$

Solution. Using product rule,

$$\begin{aligned} y' &= \frac{d}{dt}\left(e^{t^2}\right) \int_0^t e^{-s^2} ds + e^{t^2}\left(\frac{d}{dt}\int_0^t e^{-s^2} ds\right) + 2te^{t^2} \\ &= 2te^{t^2} \int_0^t e^{-s^2} ds + e^{t^2}(e^{-t^2}) + 2te^{t^2} = 2(t)\left(e^{t^2}\int_0^t e^{-s^2} ds + e^{t^2}\right) + 1 = 2ty + 1. \end{aligned}$$

Therefore,

$$\boxed{y' - 2ty = 2ty + 1 - 2ty = 1.}$$

□

3 2.1

Question 10. Find the solution of the given initial value problem:

$$y' + 2y = te^{-2t}, \quad y(1) = 0$$

Solution. Using the general form, we get that $p(t) = 2$ and $g(t) = te^{-2t}$. Therefore, the integrating factor $\mu(t)$ is

$$\mu(t) = \exp \int p(t)dt = e^{2t}.$$

Substituting this back into the original equation and multiplying both sides by $\mu(t)$, we get

$$\frac{d}{dt}\left(e^{2t}y\right) = e^{2t}(te^{-2t}) = t$$

Integrating both sides gives

$$e^{2t}y = \frac{t^2}{2} + C.$$

Using the initial value $y(1) = 0$,

$$e^{2(1)}(0) = \frac{1^2}{2} + C \implies C = -\frac{1}{2}.$$

Therefore,

$$e^{2t}(y) = \frac{t^2 - 1}{2} \implies \boxed{y = \left(\frac{t^2 - 1}{2}\right)e^{-2t}.}$$

□

Question 20. Find the value of y_0 for which the solution of the initial value problem

$$y' - y = 1 + 3\sin t, \quad y(0) = y_0$$

remains finite as $t \to \infty$.

Solution. Using the general form, we get that $p(t) = -1$ and $1 + 3\sin t$. Therefore, the integrating factor $\mu(t)$ is

$$\mu(t) = \exp \int p(t)dt = e^{-t}.$$

Substituting this back into the original equation and multiplying both sides by $\mu(t)$, we get

$$\frac{d}{dt}\left(e^{-t}y\right) = (1 + 3\sin t)e^{-t}.$$

Integrating both sides gives

$$\begin{aligned} e^{-t}y &= \int (1 + 3\sin t)e^{-t}dt = \int e^{-t}dt + 3\int \sin t e^{-t}dt \\ &= -e^{-t} + \frac{3e^{-t}(-\sin t - \cos t)}{2} + C. \end{aligned}$$

Therefore,

$$y = -1 + \frac{-3(\sin t + \cos t)}{2} + Ce^{t}.$$

When substituting $y(0) = y_0$, we get

$$y_0 = -1 - \frac{3}{2} + C \implies C = y_0 + \frac{5}{2},$$

giving us the general equation

$$y = -1 + \frac{-3(\sin t + \cos t)}{2} + \left(y_0 + \frac{5}{2}\right)e^{t}.$$

So when $t \to \infty$, the exponential term is at risk of blowing up. By setting $\boxed{y_0 = -\frac{5}{2}}$, the solutions to the problem remain finite. □

Question 23. Show that if a and λ are positive constants, and b is any real number, then every solution of the equation

$$y' + ay = be^{-\lambda t}$$

has the property that $y \to 0$ as $t \to \infty$. *Hint:* Consider the cases $a = \lambda$ and $a \neq \lambda$ separately.

Solution.

$$\mu(t) = \exp \int p(t)dt = Ce^{at},$$

$$\frac{d}{dt}e^{at}y = be^{(a-\lambda)t}$$

Integerating both sides gives us depends on two cases of when $a = \lambda$ or when $a \neq \lambda$. When $a = \lambda$, then the right hand side reduces to b. Therefore,

$$e^{at}y = bt + C \implies \boxed{y = (bt + C)e^{-at}}.$$

As $t \to \infty$, the decay exponential forces y to go to zero. Now, if $a \neq \lambda$,

$$e^{at}y = \frac{b}{a-\lambda}e^{(a-\lambda)t} \implies \boxed{y = \frac{b}{a-\lambda}e^{-\lambda t}.}$$

Similarly, the presence of the exponential decay forces $y \to 0$ as $t \to \infty$. $\square$

4 2.2

Question 9. Find the solution of the given initial value problem in explicit form and determine (at least approximately) the interval in which the solution is defined.

$$y' = (1 - 2x)y^2, \quad y(0) = -1/6$$

Solution. By separation of variables,

$$\frac{dy}{y^2} = (1 - 2x)dx.$$

Integrating both sides gives us

$$\int \frac{dy}{y^2} = \int (1 - 2x)dx \implies -\frac{1}{y} = x - x^2 + C.$$

Now, by substituting the initial value $y(0) = -1/6$,

$$-\frac{1}{-1/6} = 0 - 0^2 + C \implies C = 6.$$

Therefore, by substituting C and rearranging the equation in terms of y,

$$\boxed{y = \frac{1}{x^2 - x - 6}.}$$

We can factor the denominator to be $(x-3)(x+2)$. Since the initial value starts from $x = 0$, the interval in which the solution is defined is $\boxed{(-2, 3)}$ □

Question 11. Find the solution of the given initial value problem in explicit form and determine (at least approximately) the interval in which the solution is defined.

$$x\,dx + ye^{-x}\,dy = 0, \quad y(0) = 1$$

Solution. By separation of variables,

$$xe^x\,dx = -y\,dy.$$

Integrating both sides gives us

$$\int xe^x\,dx = -\int y\,dy \implies xe^x - e^x = -\frac{y^2}{2} + C.$$

Substituting the initial value $y(0) = 1$,

$$0 \cdot e^0 - e^0 = -\frac{1}{2} + C \implies C = -\frac{1}{2}.$$

Therefore, by substituting C and rearranging in terms of y, since $y(0) = 1 > 0$, we take the positive of the square root as such:

$$\begin{aligned} xe^x - e^x &= -\frac{y^2}{2} - \frac{1}{2} \\ -2xe^x + 2e^x &= y^2 + 1 \\ &\boxed{y = \sqrt{-2xe^x + 2e^x - 1}.} \end{aligned}$$

The solution is defined on the interval where the radicand $-2xe^x + 2e^x - 1$ is non-negative and contains the initial point $x = 0$. Solving this numerically, the expression equals zero around $x \approx -1.68$ and $x \approx 0.77$, so the solution is defined for $x \in (-1.68, 0.77)$. $\square$

Question 15. Find the solution of the given initial value problem in explicit form and determine (at least approximately) the interval in which the solution is defined.

$$y' = (3x^2 - e^x)/(2y - 5), \quad y(0) = 1$$

Solution. By separation of variables,

$$(2y - 5)\, dy = (3x^2 - e^x)\, dx.$$

Integrating both sides gives us

$$\int (2y - 5)\, dy = \int (3x^2 - e^x)\, dx \implies y^2 - 5y = x^3 - e^x + C.$$

Substituting the initial value $y(0) = 1$,

$$1^2 - 5(1) = 0^3 - e^0 + C \implies -4 = -1 + C \implies C = -3.$$

Completing the square on the left-hand side,

$$\left(y - \frac{5}{2}\right)^2 = x^3 - e^x - 3 + \frac{25}{4} = x^3 - e^x + \frac{13}{4}.$$

Since $y(0) = 1 < 5/2$, we take the negative branch:

$$y = \frac{5}{2} - \sqrt{x^3 - e^x + \frac{13}{4}}.$$

The solution is defined when the radicand is non-negative, i.e., $x^3 - e^x + \frac{13}{4} \geq 0$. Solving numerically, the solution is defined approximately on

$$x \in (-1.4,\ 2.0).$$

□

Question 24. Use separation of variables to solve the differential equation

$$\frac{dQ}{dt} = r(a + bQ), \quad Q(0) = Q_0,$$

where a, b, r, and Q_0 are constants. Determine how the solution behaves as $t \to \infty$.

Solution. By separation of variables,

$$\frac{1}{a + bQ}\, dQ = r\, dt.$$

Integrating both sides gives us

$$\int \frac{1}{a + bQ}\, dQ = r \int dt \implies \frac{1}{b} \ln|a + bQ| = rt + C.$$

Substituting the initial value $Q(0) = Q_0$,

$$C = \frac{1}{b} \ln|a + bQ_0|.$$

Substituting C back and solving for Q,

$$\frac{1}{b} \ln|a + bQ| = rt + \frac{1}{b} \ln|a + bQ_0| \implies \ln|a + bQ| - \ln|a + bQ_0| = brt$$

$$\implies \ln\left|\frac{a + bQ}{a + bQ_0}\right| = brt \implies a + bQ = (a + bQ_0)e^{brt}.$$

Therefore,

$$\boxed{Q(t) = \frac{a + bQ_0}{b} e^{brt} - \frac{a}{b}.}$$

As $t \to \infty$, the behavior depends on the sign of br. If br is positive, then $Q(t) \to \infty$. If br is negative, then $e^{brt} \to 0$, so $Q(t) \to -\frac{a}{b}$. Lastly, if $br = 0$, then $Q(t) = Q_0$ remains constant. □